LEARNING JOURNEY ON SAP CPE

VOL 2

ABSTRACT

Explore the market volatility using the SAP Commodity Management using CPE. This will take you through the concept of market data & Indexes, how to handle Deals and Pricing for Commodities

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Commodity Management with CPE

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Introduction

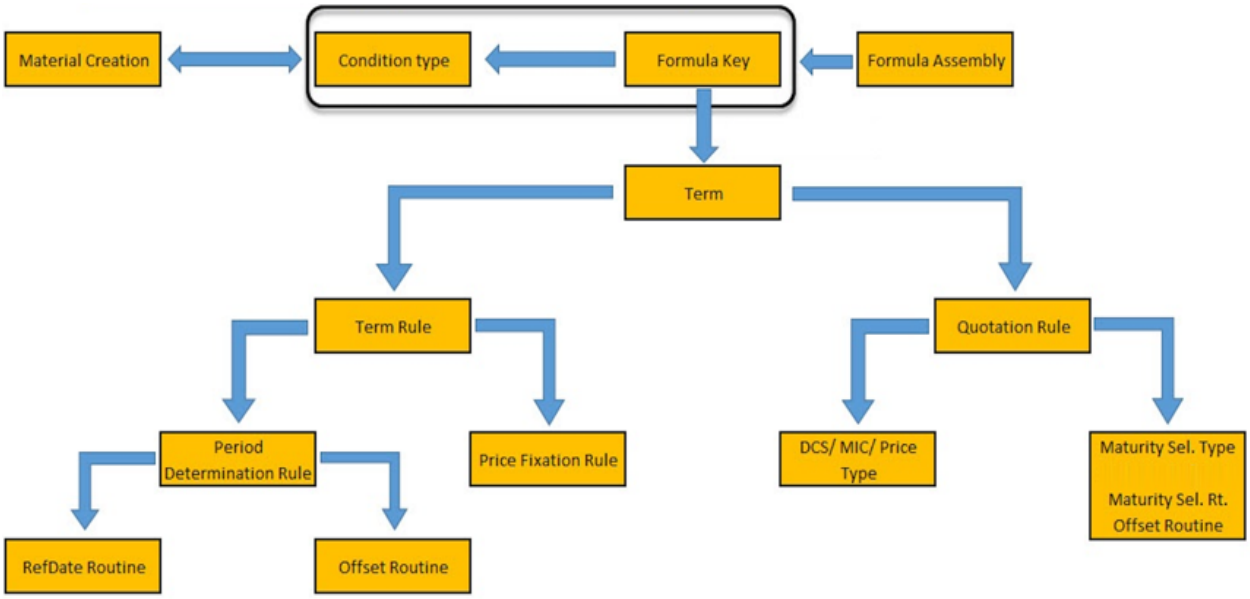
This document is a high-level overview on SAP S4 Commodity Management with Commodity Pricing Engine (CPE). The concept should work good with every other industry (like Oil and Gas, Agri or Mining Industry), but the examples used in this case will be more from Oil and Gas Industry (like Crude/ Gasoline, etc.).

This book (Volume) shall be used as a follow up learning document on top of the Volume 1, which has already been published.

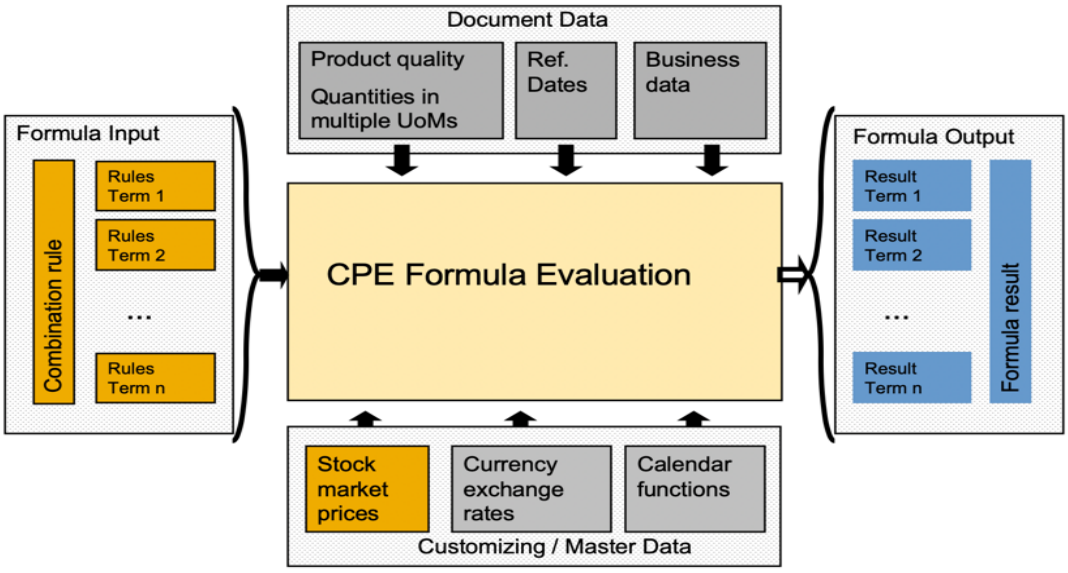
This Volume is dedicated to the high-level concept about commodities and pricing using CPE in S4. This includes the basics about the pricing based on market. It contains the deep dive for CPE. To look into the Volume 1 of this Book, you may go to the below link here.

Recap:

In Volume 1 of this book, we saw how CPE has different elements within itself, which work in conjunction to provide the final formula rate. CPE contains too may data objects and Routines which work together to calculate the market prices and land on a settlement and mark to market price for a deal (contract). At a high level, a CPE structure looks like below.



Architecture of CPE objects work together to determine the final output price (settlement price and/ or Mark to Market Price).



Now we shall see the important Routines within CPE that can play a vital role. Also, the role of BRFPLUS module in running the Commodity Pricing Engine

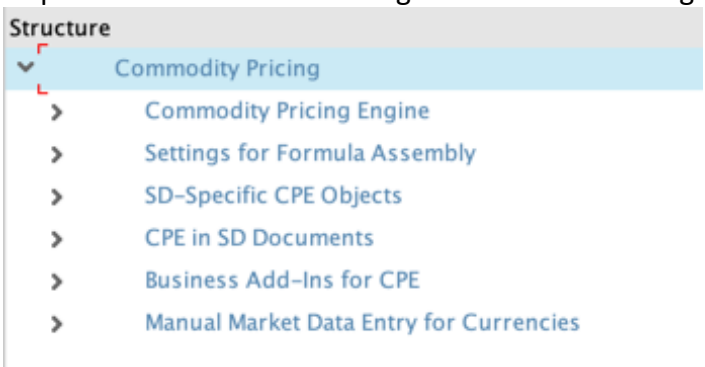
CPE Customizing

Some of the most important and commonly used SAP Transaction Codes for CPE customizing are:
VCPE_WB or MCPE_WB
VCPE_RULE or MCPE_RULE
VCPE_CUS or MCPE_CUS
VCPE_MD or MCPE_MD

Where, V stands for Sales side and M stands for Purchase side set up. Most of the customizing objects within CPE space, are independent of application (V or M), and can be accessed from either of the transactions. Like DCS/ MIC/ PT or Formula, etc.

The T Code used for setting up the CPE routines, is VCPE_CUS (or MCPE_CUS). I will be using the application V, to show the different pricing objects. The behavior/ set up will be similar/ same for the M side of the application.

The screenshots in this document are taken from internal SAP system, and is not intended to display any functionality, related to any of the clients of SAP. Any resemblance would be purely co-incidental. Most of the standard functionality might fit many of the customer's requirements. I will be showing the standard offerings from SAP.



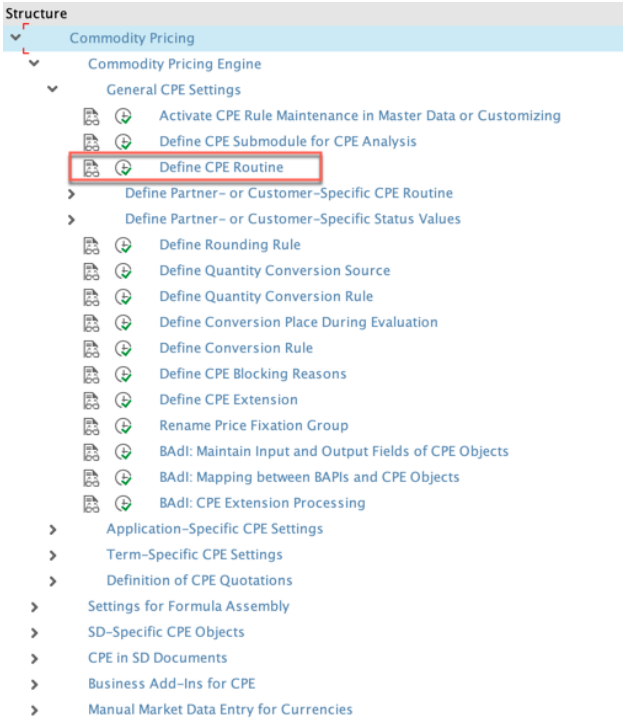
PIC TAKEN FROM INTERNAL SAP SYSTEM

We shall be focusing on

a) CPE Routines and
b) Important set up for CPE, *which is not covered in Volume 1 of this book*
c) BRF+ relevancy for CPE pricing

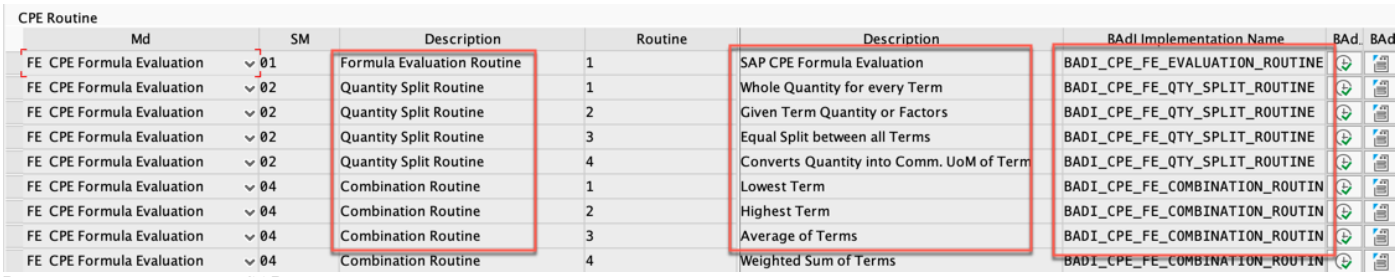
CPE Routines

The CPE Routines can be viewed under the tree node shown below. Here, you can see all the available Routines within each sub-module. You can also create your custom routine, to meet your customer's requirement, by having a BADi implementation. You can see the available BADi implementations, which can be enhanced.



PIC TAKEN FROM INTERNAL SAP SYSTEM

Here you can see the routines that work within the different tree elements of CPE (see tree diagram above). These are standard delivered. The Sub-Module tells the intent of the Routine (BAdi). The names are self-explanatory, and you can find them in one of the elements within CPE.



CPE Routine

Md	SM	Description	Routine	Description	BAdI Implementation Name
FE CPE Formula Evaluation	01	Formula Evaluation Routine	1	SAP CPE Formula Evaluation	BADI_CPE_FE_EVALUATION_ROUTINE
FE CPE Formula Evaluation	02	Quantity Split Routine	1	Whole Quantity for every Term	BADI_CPE_FE_QTY_SPLIT_ROUTINE
FE CPE Formula Evaluation	02	Quantity Split Routine	2	Given Term Quantity or Factors	BADI_CPE_FE_QTY_SPLIT_ROUTINE
FE CPE Formula Evaluation	02	Quantity Split Routine	3	Equal Split between all Terms	BADI_CPE_FE_QTY_SPLIT_ROUTINE
FE CPE Formula Evaluation	02	Quantity Split Routine	4	Converts Quantity into Comm. UoM of Term	BADI_CPE_FE_QTY_SPLIT_ROUTINE
FE CPE Formula Evaluation	04	Combination Routine	1	Lowest Term	BADI_CPE_FE_COMBINATION_ROUTIN
FE CPE Formula Evaluation	04	Combination Routine	2	Highest Term	BADI_CPE_FE_COMBINATION_ROUTIN
FE CPE Formula Evaluation	04	Combination Routine	3	Average of Terms	BADI_CPE_FE_COMBINATION_ROUTIN
FE CPE Formula Evaluation	04	Combination Routine	4	Weighted Sum of Terms	BADI_CPE_FE_COMBINATION_ROUTIN

PIC TAKEN FROM INTERNAL SAP SYSTEM

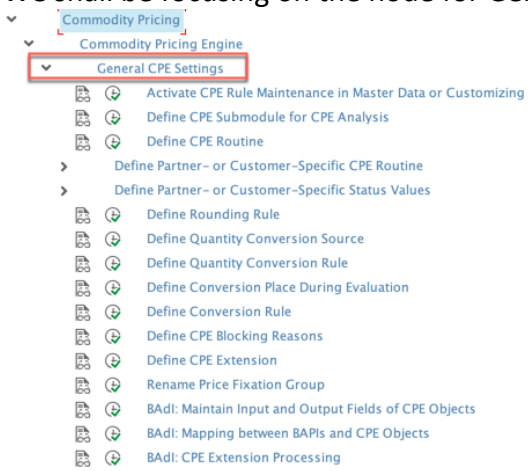
For an example, you can see different routines available for Formula Evaluation, Combination Routine, Quantity Split Routine (used in Formula Key), etc. Beside to each Sub-Module, you can see the delivered BAdi implementation, which can be used to create any custom enhancements.

Some of the important sub-modules that you can find are namely Combination Routine, Quantity Split Routine, Closed Days Routine, Period Determination Routine, Date Offset Routine, Quantity Conversion Routine, etc. The ones mentioned are most commonly used to create custom routines. You can find these routines in one of the CPE elements used for pricing.

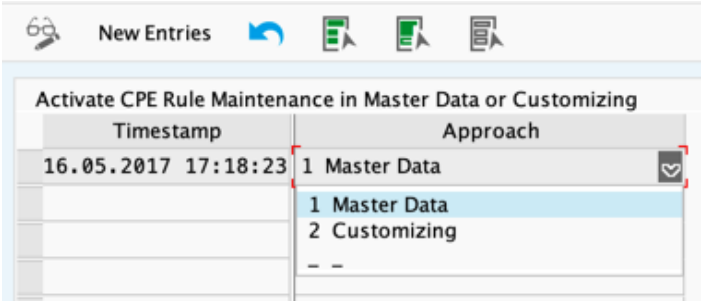
Important CPE Set up

This section is used to cover the important customizing tools used within CPE that are run by default, to calculate the market prices.

We shall be focusing on the node for General CPE settings:



Activating the CPE Rule maintenance as Master Data or Customizing. Using this you can define if you would like the objects like Formula, Term Key, to be able to maintain as a master data, or as config item only.



Quantity Conversion Rule. There are some standard delivered conversion rules for quantity conversions. This is mostly used when the quoted price and the transaction volume do not share the same UoM. In that case there will be a quantity conversion done to calculate the correct rate/ price.

These rules have conversion routines assigned to it, which runs in the background to do the conversions. You can create your own custom routine to meet any specific requirement. For that you may have to create your new conversion rule.

CPE – Quantity Conversion Rule

Rule for Qu...	'Valid From' Time Stamp	Description	Quantity C...	Conversion Rule Description	Rounding Rule for Quantity...
10	01.01.2000 00:00:00	Material Master	1	Material Master	
20	01.01.2000 00:00:00	Manual Factors	2	Manual Factors	
40	01.01.2000 00:00:00	Manual Factors from ERP Pricing	4	Manual Factors from ERP Pricing	
60	01.01.2000 00:00:00	Adjusted Quantity	6	Adjusted Quantity	
110	01.01.2000 00:00:00	Order: Material Master; Billing: Manual	11	Mat. Master orders; man. factors Invo...	
120	01.01.2000 00:00:00	Order: Material Master; Billing: Batch	12	Mat. Master orders; Batch Invoice	
150	01.01.2000 00:00:00	Batch if entered else Material Master	15	Order with Batch	
160	01.01.2000 00:00:00	Order/Return: Mat. Mstr; Billing: Batch	16	Order/Return: Mat. Mstr; Billing: Batch	

To each conversion rule, you can assign a rounding rule, to define the decimal places for the calculated quantity.

Conversion place. There are a lot of conversion that happen during the process of CPE evaluation of a Formula. Some of the examples that can be outlined, are, converting the FX rates, converting the quantity conversions, converting the quotation rates or the surcharge rates. These conversions can happen at Term level or to the Formula level, etc. These specific spots, where these conversions can happen, are called conversion places. We might not fall into a need for any custom conversion places.
These can be pure conversion of value, or may be the rounding of the calculated values.

CPE Conversion Place

Conve...	Description	Conv. Category
0	Generic	2 Rounding
11	Evaluation: Commodity UoM to Formula UoM	1 Conversion
12	Evaluation: Formula to Document Currency	1 Conversion
13	Evaluation: Gross/Net Subst. Formula UoM	1 Conversion
21	Quantity Split Routine: Round Term Quan.	2 Rounding
22	QtySplit R: FormCommQty into TermCommUoM	1 Conversion
31	Combination Rule: Term to Formula Rate	1 Conversion
32	Combination Rule: Round Formula Rate	2 Rounding
33	Combination Rule: Comm.Qty into Term UoM	1 Conversion
41	Combination Routine: Round Group Result	2 Rounding
61	Source Routine: Quotation to Quot.Grp.	1 Conversion
62	Calculation Rule: Quotation <-> TermRate	1 Conversion
63	Calculation Rule: Limit to Term Rate	1 Conversion
64	Calculation Rule: Round Result	2 Rounding
71	Calculation Routine: Round Interm. Rate	2 Rounding
72	Calculation Routine: Round Calcul. Rate	2 Rounding
81	Surcharge Rule: Gross/Net Subst. Surch.	1 Conversion
82	Surcharge Rule: Surcharge <-> Term Rate	1 Conversion
83	Surcharge Rule: Round Surcharged Rate	2 Rounding
91	Surcharge Routine: Round Interm. Result	2 Rounding
101	Period Determ.: Gross/Net Subst. Quot.	1 Conversion
171	Source Routine: Round Selected Quotation	2 Rounding
172	Source Routine: Round Selected Rate	2 Rounding
181	Round Averaged Exchange Rate	2 Rounding

During the execution of the code, these conversion places are run automatically, and they do the conversions, based on the standard delivered set up. If there is any change required, the user can adjust that, by assigning a new conversion rule to such conversion places at the Formula or Term level.

Conversion Rule**.** There are some standard delivered conversion rules, which controls how the conversion of Term or Quotation values (for an example) would happen while Formula Evaluation. These conversion rules have Conversion Routines assigned to itself. All these routines can be found in the 'General CPE Routines' section, covered in the earlier section of the document.
If there is any requirement to change the behavior of any conversion rule, a new routine can be created and assigned to a new conversion rule. It is recommended not to change the standard rules, but to create any new custom data.
Quantity conversion rules can be assigned to these rules which control the conversion of a volume among different UoM(s).

CPE – Conversion Rule

Rule for Co..	'Valid From' Time Stamp	DefltConv	Description	ConvRtn	ExRt	Av..	Rounding R..	Rule for Qu..
110	.01.2000 00:00:00	✓	First Material Master, then Rate Type M	1	M			10
111	01.01.2000 00:00:00		Billing: Manual, then Rate Type M	1	M			110
112	01.01.2000 00:00:00		Billing: Batch, then Rate Type M	1	M			120
113	01.01.2000 00:00:00		Billing: QCI, then Rate Type M	1	M			130
120	01.01.2000 00:00:00		First Manual Factors, then Rate Type M	1	M			20
160	01.01.2000 00:00:00		Adjusted Quantity, Rate Type M	1	M			60
210	01.01.2000 00:00:00		First Rate Type M, then Material Master	2	M			10
211	01.01.2000 00:00:00		First Rate Type M, Billing: Manual	2	M			110
212	01.01.2000 00:00:00		First Rate Type M, Billing: Batch	2	M			120
213	01.01.2000 00:00:00		First Rate Type M, Billing: QCI	2	M			130
220	01.01.2000 00:00:00		First Rate Type M, then Manual Factors	2	M			20
1000	01.01.2000 00:00:00		Currency Conversion with Rate Type M	10	M			
2010	01.01.2000 00:00:00		Quantity Conversion with Material Master	20				10
2011	01.01.2000 00:00:00		Quantity Conversion, in Billing: Manual	20				110
2012	01.01.2000 00:00:00		Quantity Conversion, in Billing: Batch	20				120
2013	01.01.2000 00:00:00		Quantity Conversion, in Billing: QCI	20				130
2020	01.01.2000 00:00:00		Quantity Conversion with Manual Factors	20				20

During the execution of the code, these conversion places are run automatically, and they do the conversions, based on the standard delivered set up. If there is any change required, the user can adjust that, by assigning a new conversion rule to such conversion places at the Formula or Term level.

CPE Blocking Reasons. There are standard delivered blocking reasons for any sales order/ purchase order. These blocking reasons are different from the core SD blocking reasons for sales orders.
These blocking reasons are used to control the orders from invoicing, depending on the pricing status. As you can see from the screenshot below, these speak more from the status of market prices. If the market price or an FX rate, is provisional or missing, we can put a blocking status at the Term/ Formula level, to block it from creating final invoice. In case of Provision block, you can still go ahead and create a Provisional Invoice.

CPE Billing: Blocking Reason to be set for Document Items

CPE Block	Description
0C	Missing Currency Exchange Rate
0P	Provisional Price
0Q	Missing Price Quotation

If you want to ignore such blocks for any specific condition type, you can do so, using the node: Define Condition-Type-Specific Settings under Application-Specific CPE settings.
Also, you can assign blocking reasons for Billing documents and assign the CPE blocking reasons the these blocking reasons (codes) to get them triggered for a certail Billing Document type.

Condition-Type-Specific CPE Settings

Appl	CTy.	'Valid From' Time Stamp	No Blocks
M ERP Material...	MBSY	01.01.2019 00:00:00	☑
M ERP Material...	MP2C	01.01.2019 00:00:00	☑
M ERP Material...	MP2G	01.01.2019 00:00:00	☑
M ERP Material...	MP2S	01.01.2019 00:00:00	☑
M ERP Material...	MPSY	01.01.2019 00:00:00	☑
V ERP Sales an...	CMP9		☑
V ERP Sales an...	MBSY	01.01.2019 00:00:00	☑
V ERP Sales an...	MP2C	01.01.2019 00:00:00	☑
V ERP Sales an...	MP2G	01.01.2019 00:00:00	☑
V ERP Sales an...	MP2S	01.01.2019 00:00:00	☑
V ERP Sales an...	MPCL	01.06.2019 14:37:49	☑
V ERP Sales an...	MPSY	01.01.2019 00:00:00	☑

Structure

- Commodity Pricing
  - Commodity Pricing Engine
  - Settings for Formula Assembly
  - SD-Specific CPE Objects
  - CPE in SD Documents
    - Define Blocking Reason For Billing
    - Assign CPE Blocking Reasons to SD Blocking Reasons
    - Define Billing Types of Diff. Invoicing for Billing Due List
  - Business Add-Ins for CPE
  - Manual Market Data Entry for Currencies

BRF+ relevancy for CPE pricing

Settings for Formula Assembly. This setting is required to set up the trigger of Formula Assembly Evaluation from BRF+. BRF+ has the Decision table which control the Formula and Term level details to be used for Formula Evaluation.

This document does not cover the details on the BRF+ working model but will focus on how to call the BRF+ from CPE.

The BRF+ standard delivered Application for CPE, has pre-existing data elements, which play a vital role in determining the right Formula, etc. If you need any new data elements, as a part of BRF+ determination logic, you can add them to the list.

Structure

- Commodity Pricing
  - Commodity Pricing Engine
  - Settings for Formula Assembly
    - Maintain Field Catalog
    - Maintain Field Mapping
    - Assign BRFplus Application to Pricing Procedure
    - BAdI: Formula Assembly for Commodity Pricing Engine
  - SD-Specific CPE Objects
  - CPE in SD Documents
  - Business Add-Ins for CPE
  - Manual Market Data Entry for Currencies

Maintain field Catalog allows you to add any new field catalog (data elements) to the BRF+ structures. This addition is done on the CPE side for the Formula Assembly. You will have to add same data elements in the BRF+ module as well.

CPE FA Fieldattributes

Formula Assembly Fieldname	Data element
CONTRACT	VGBEL
CURRENCY	WAERK
CUST_GROUP	KDGRP
DIS_CHANNEL	VTWEG
DIVISION	SPART
DOCUMENT	BELNR_D
INCO1	INCO1
INCO2	INCO2
INCO2_L	INCO2_L
INCO3_L	INCO3_L
INFO_TYPE	ESOKZ

Maintain Field Mapping. Then you will have to map the field of the Formula Assembly to the S4 structures, like the pricing structure, for an example. This will help drive the right field mapping between S4 and BRF+ fields.
You must maintain such mappings for each Application you would want the determination logic to work, based on the fields.

Application: V Sales and Distribution

CPE FA Fieldmapping

Structure name	Component name	Formula Assembly Fieldname
KOMK	INCO1	INCO1
KOMK	INCO2	
KOMK	INCO2_L	INCO2_L
KOMK	INCO3_L	INCO3_L
KOMK	KDGRP	CUST_GROUP
KOMK	KUNNR	SOLD_TO_PARTY
KOMK	KUNWE	SHIP_TO_PARTY
KOMK	SPART	DIVISION
KOMK	VKORG	SALES_ORG
KOMK	VTWEG	DIS_CHANNEL
KOMP	KONDM	MAT_PR_GROUP
KOMP	KPOSN	REFITEM
KOMP	MATNR	MATERIAL
KOMP	VGBEL	CONTRACT
KOMP	WERKS	PLANT
VBAK	VBELN	CONTRACT
VBAP	ROUTE	ROUTE

Assign BRFplus Application to Pricing Procedure. At the end, you must assign a BRF+ Application/ Function to the Pricing Procedure, for each Application. This will trigger the BRF+ to use the right rules to determine the business logic, to derive the right Formula & Term level details to be used in any document for CPE evaluation.

New Entries

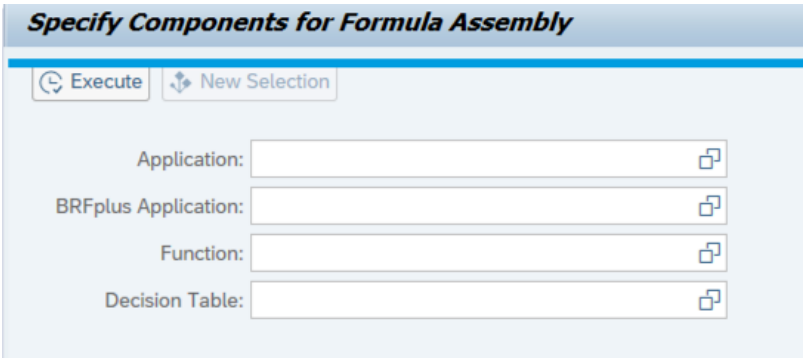
Dialog Structure
- Assign BRF+ Application
- Assign BRF+ Functions

Application: V Sales and Distribution
Procedure: CMAL01
BRFplus Appl. Name: COPPER_CPE_FA_DEVTEST

Assign BRF+ Functions

BRFplus Function
APPL_V_FUNCTION_TERM
APPL_V_FUNCTION_FORMULA

Specify Components for Formula Assembly. Using the Transaction VCPE_MD (for application V) you can assign the Business Logic/ data to derive the Formula/ Term Level Details. For purchase side, it will be MCPE_MD.

Structure
- Commodity Pricing
  - Specify Components for Formula Assembly
  - Specify CPE Quotations and Formulas
  - Edit Commodity Prices
- Basis Pricing
  - Commodity Curve Maintenance

Here, you can assign the Application (e.g. V or M) BRF+ Application, Function and Decision table names to enter the Business data.